Amplification and Time-Resolution of Weak Ultrafast Light Pulses

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Introduction

Statement of Problem: In spectroscopy, sensing, and light-matter experiments, there is a need to characterize weak, complicated, broadband, ultrashort light pulses.

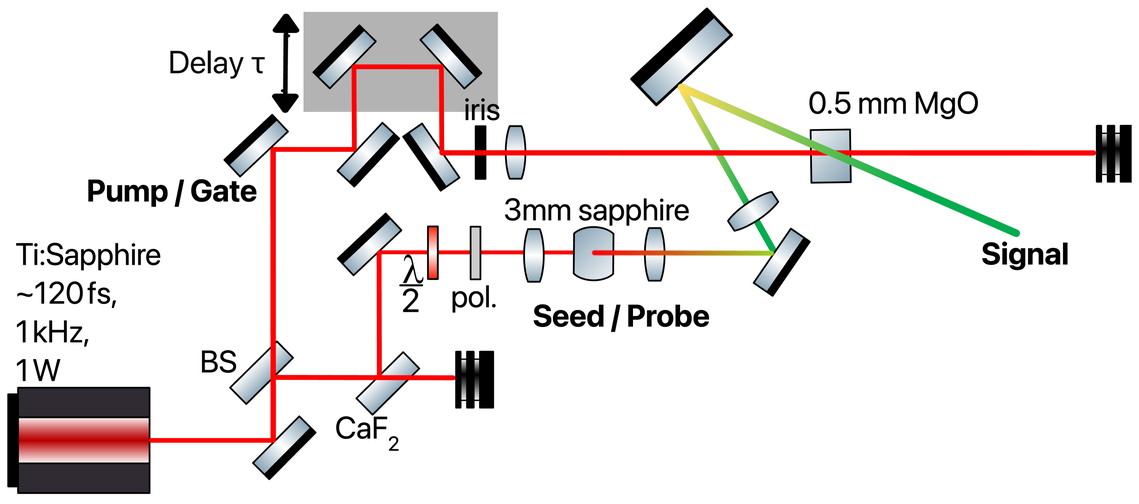
Motivation: When photon intermodulation ("wave mixing") is taken to the extreme case of laser intensities in the TW/cm^2 regime, annihilation and amplification of the incident photons ensues.

Goals: Harness Kerr-instability amplification (KIA) to **amplify** supercontinuum spectra. Use cross-correlation frequency-resolved optical gating (X-FROG) algorithm to **reconstruct** the original spectrum, pulse profile, and phase of light.

Experiment

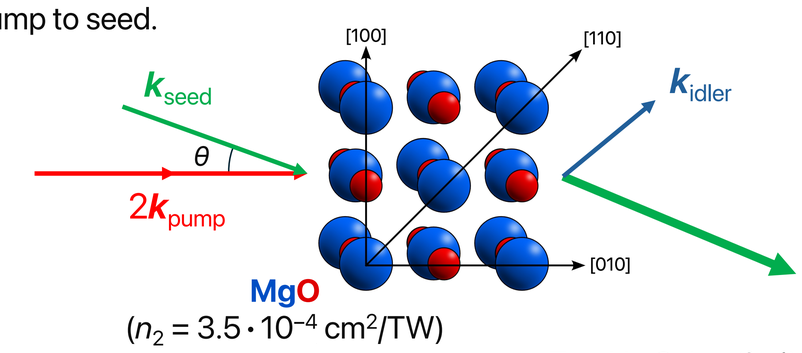
Spatiotemporally overlap pump and supercontinuum seed beams in MgO.

Create a **spectrogram** by measuring amplified spectra while interferometrically delaying pump-seed overlap.



Theory of Kerr-Instability Amplification

Mixing of pump and seed beams in a nonlinear optical crystal allows for energy transfer from pump to seed.



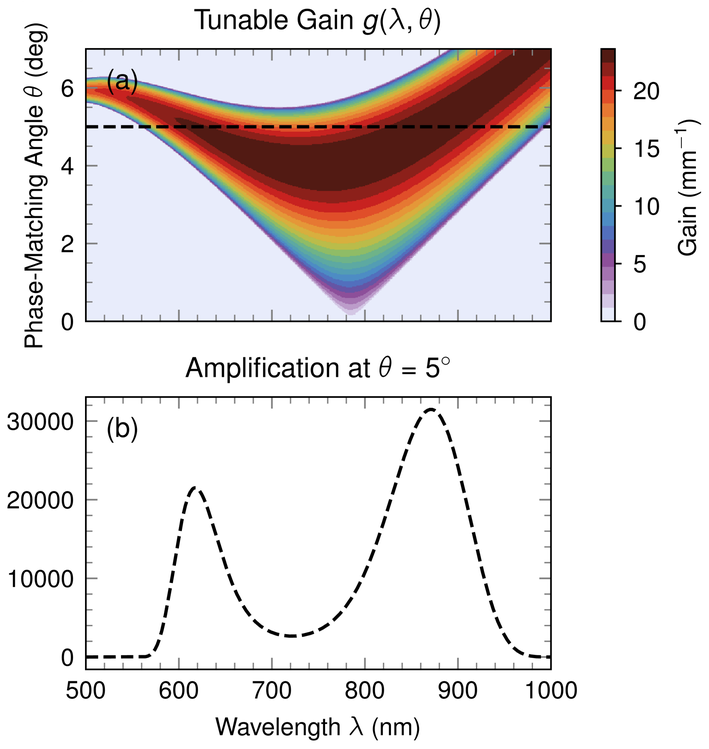
($n_2 = 3.5 \cdot 10^{-4}$ cm^2/TW)

$$g = \frac{\omega_p}{cn(\omega_p)} n_2 I_{pump}(t)$$

$$I_{seed}(L) = I_{seed}(0)\cosh^2(\tfrac{1}{2} gL)$$

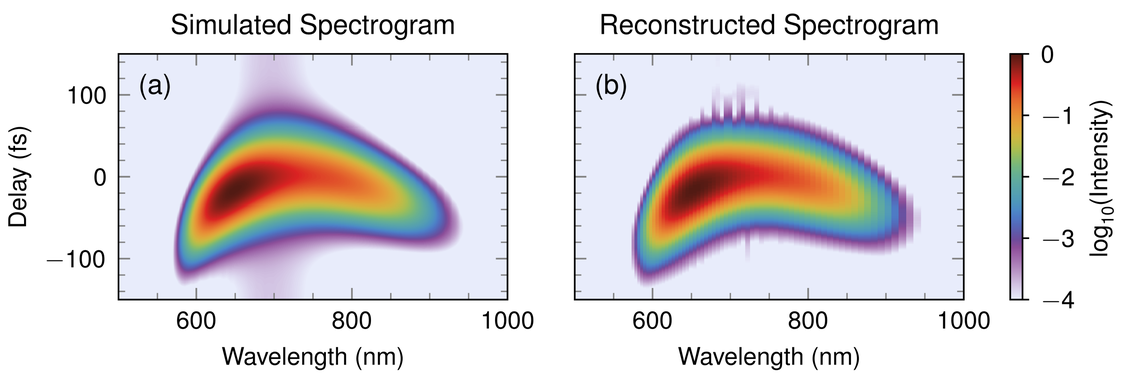
Highly **tunable** gain mechanism with **exponential** sensitivity to the pump intensity!

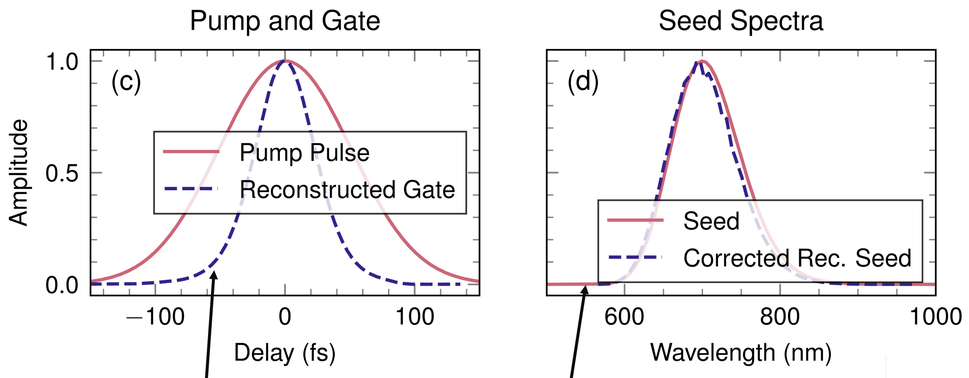
In MgO, a pump intensity of 15 TW/cm^2 can amplify seed pulses by more than **10 000 times**.



Results

Simulate spectrogram **construction** using KIA gate; **reconstruct** iteratively with X-FROG

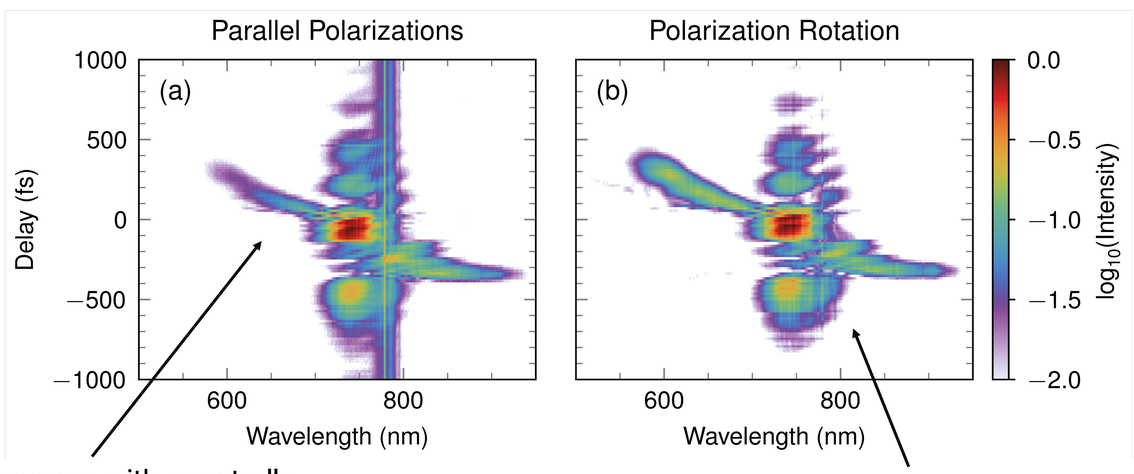




Effective gate is narrower than the laser pump, which enables better time resolution

Corrected reconstructed seed reproduces original seed

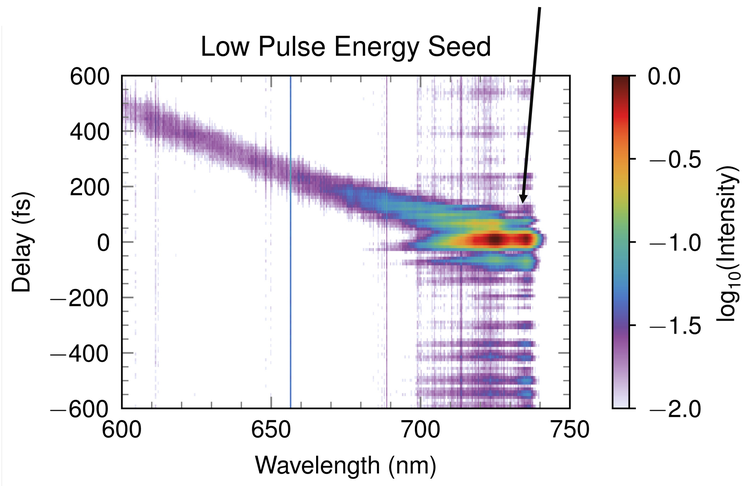
Can gate the polarization of the seed to cause **polarization rotation** in the crystal



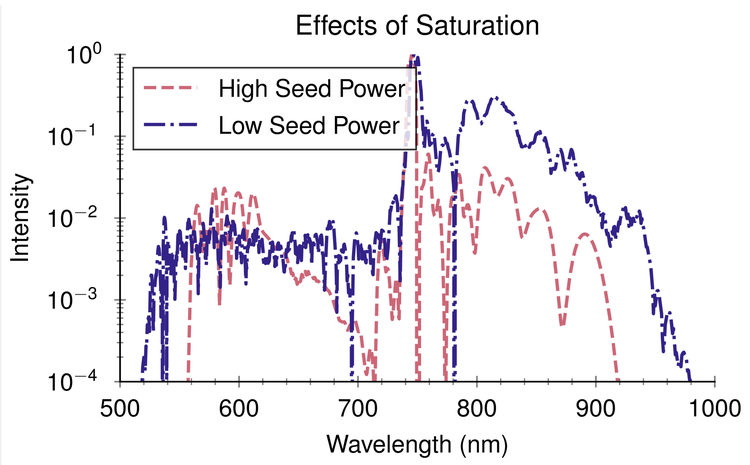
Measured spectrogram with spectrally broad supercontinuum seed

Remove seed background with simultaneous polarization gating

Can amplify **extremely weak** pulses of about 100 aJ



Can **saturate** gain process



Takeaway: Can reconstruct nearly an octave (500 nm – 1000 nm) of a complicated, **chirped** pulse.

Conclusions

- Can amplify pulses spanning **nearly an octave** down to the hundred attojoule regime—corresponding to mere **hundreds of photons**.
- Gain mechanism is **tunable**, can be made **background-free**, and has demonstrated **saturation** effects.
- Introduced "Kerr-instability cross-correlation induced nonlinear gate frequency-resolved optical gating" (**KICKING FROG**).

Future Work

- **Spectroscopy**: Use KICKING FROG as a **characterization tool** for pump-probe spectroscopy.
- **Plasma Dynamics**: Measure **plasma emission** wrought by KIA.
- **Quantum Optics**: Amplify and time-resolve **bright-squeezed vacuum** with KICKING FROG.

Read the preprint!